

**DATABASE SYSTEMS ENGINEERING AND
DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No - 8	Team No - 21	Project Title - Cross-Platform Content Item App for Media & Entertainment
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1. Abstract

The **Content Item Management & Analytics Portal for Media & Entertainment** is a web-based platform designed to digitize and streamline the management of digital media content. Media and entertainment organizations often handle a large number of content items such as movies, web series, television programs, trailers, and other digital assets, making it difficult to efficiently organize, track, and analyze content using traditional methods.

The proposed system provides a centralized platform for managing content items throughout their lifecycle, from creation and review to publication and archival. Subscribers can register and access content according to their assigned roles and permissions. The system provides powerful search and filtering capabilities to help users quickly discover relevant content. A recommendation feature can further assist subscribers in discovering content based on available content information and user activity.

The portal also maintains notifications and audit logs to track important activities and keep users informed about relevant updates. An analytics dashboard provides management with visual insights into content distribution, user activity, content status, and other relevant statistics. Overall, the system aims to improve content organization, accessibility, operational monitoring, and data-driven decision-making within media and entertainment operations.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030004	Vedhanth	
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3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

S.No.	Category	Framework / Platform / Software	Purpose
1	Frontend Framework	React.js	Used to develop the interactive and component-based user interface.
2	Build Tool	Vite	Used to create, develop, and build the React frontend application.
3	Programming Language	JavaScript (ES6+)	Used for frontend and backend application logic.
4	Markup Language	HTML5	Used to structure the web application pages.
5	Styling	CSS3	Used for styling, responsive layouts, animations, themes, and the cinematic UI.
6	Routing	React Router	Used to navigate between different pages and manage protected routes.
7	HTTP Client	Axios	Used for communication between the React frontend and backend REST APIs.
8	UI Icons	Lucide React	Used for navigation, buttons, menus, dashboards, and other interface icons.
9	Data Visualization	Recharts	Used to create charts for movie, revenue, rating, user, and genre analytics.
10	Backend Runtime	Node.js	Provides the runtime environment for executing the backend application.
11	Backend Framework	Express.js	Used to develop REST APIs and handle HTTP requests and responses.
12	Database	MySQL	Used to store and manage users, movies, genres, reviews, ratings, favorites, watchlists, and other application data.
13	Database Driver	MySQL2	Used to establish communication between the Node.js backend and MySQL database.
14	Authentication	JSON Web Token (JWT)	Used for secure user authentication and protected access to application resources.
15	Password Security	bcrypt	Used to securely hash and verify user passwords.
16	Security	Helmet.js	Used to add security-related HTTP headers to the Express application.
17	Cross-Origin Resource Sharing	CORS	Used to allow secure communication between the frontend and backend running on different origins.
18	API Testing	Postman	Used to develop, test, and verify backend REST API endpoints.
19	Code Editor / IDE	Visual Studio Code	Used for developing and managing the frontend and backend source code.
20	Database Management	MySQL Workbench	Used to create, manage, query, and test the MySQL database.
21	Database IDE	DataGrip	Used as an alternative database development and management environment for MySQL.

S.No.	Category	Framework / Platform / Software	Purpose
22	UI/UX Design	Google Stitch	Used to design the application's screens and establish the Letterboxd-inspired visual design.
23	Version Control	Git	Used to track source-code changes and manage different versions of the project.
24	Repository Hosting	GitHub	Used to store the source code and manage the project's version history.

Guide Name and Signature (remarks if any) _____